Activity 05 - Compare Pretrained Google Cloud AI APIs

Course: Google Professional Machine Learning Engineer Training Course code: C740 Version: v2.0 | 12 September 2026 Evidence: activity artifacts

Objective

Prepare Vision, Document AI and Natural Language requests and interpret mock responses.

Deliverables

- vision_request.json
- mock_vision_response.json
- api_selector.py
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables, screenshots without secrets, and a short decision note. Use filenames that include the activity number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 - Inspect vision_request.json and identify the feature type, image source and maximum results

Inspect vision_request.json and identify the feature type, image source and maximum results.

```
python3 -m json.tool vision_request.json
```

Check: Record the observable result or explain the error before continuing.

Step 2 - Inspect mock_vision_response.json; list detected labels, confidence scores and the response field holding errors

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Check: Record the observable result or explain the error before continuing.

Step 3 - Run the selector for three scenarios and record the recommended managed API or custom-model path

Run the selector for three scenarios and record the recommended managed API or custom-model path.

```
python3 api_selector.py
```

Check: Record the observable result or explain the error before continuing.

Step 4 - Prepare an authenticated REST command using gcloud auth print-access-token; do not store the token

Prepare an authenticated REST command using gcloud auth print-access-token; do not store the token.

Check: Record the observable result or explain the error before continuing.

Step 5 - If the trainer approves live use, enable the chosen API and send only the supplied non-sensitive sample

If the trainer approves live use, enable the chosen API and send only the supplied non-sensitive sample.

Check: Record the observable result or explain the error before continuing.

Step 6 - Interpret 403, 429 and low-confidence responses separately: authorization, quota and model fit are different problems

Interpret 403, 429 and low-confidence responses separately: authorization, quota and model fit are different problems.

Check: Record the observable result or explain the error before continuing.

Step 7 - Design a confidence-based human-review queue and calculate the expected review rate from the mock response set

Design a confidence-based human-review queue and calculate the expected review rate from the mock response set.

Check: Record the observable result or explain the error before continuing.

Step 8 - Save the request, redacted response, operation ID, cost estimate and decision note

Save the request, redacted response, operation ID, cost estimate and decision note.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

[ ] Every required artifact exists and opens without error.

[ ] Results are tied to the supplied mock data or an approved training project.

[ ] The conclusion cites metrics, IDs or command output rather than opinion alone.

[ ] No credentials, tokens, personal data or unredacted sensitive prompts are present.

[ ] Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work, use the Console resource views and activity evidence to verify deletion. Do not delete a shared project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.